Lab6.4-Topic-classification-BERT

March 16, 2025

1 Lab6.4-Topic-classification-BERT

In this notebook, we demonstrate how to fine-tune BERT for topic classification.

We will use the simpletransformers library: wrapper for the huggingface transformers library on PyTorch.

We are going to run the notebook on colab, which has (limited) free access to GPUs.

You need to enable GPUs for the notebook:

- navigate to Edit → Notebook Settings
- select GPU from the Hardware Accelerator drop-down

1.0.1 Install/import libraries

Install the simpletransformers library (restart your runtime after the installation)

[ ]:
```
!pip install simpletransformers --upgrade
```

[2]:
```
# Import libraries
import pandas as pd
import numpy as np
import sklearn
from sklearn.metrics import classification_report
from simpletransformers.classification import ClassificationModel,␣
 ↪ClassificationArgs
import matplotlib.pyplot as plt
import seaborn as sn

# pip install tokenizers==0.20.0
```

Import the 20 newsgroups text dataset.

The dataset contains around 18,000 newsgroups posts on 20 topics.

[3]:
```
from sklearn.datasets import fetch_20newsgroups

# load only a sub-selection of the categories (4 in our case)
categories = ['alt.atheism', 'comp.graphics', 'sci.med', 'sci.space']
```

```
# remove the headers, footers and quotes (to avoid overfitting)
newsgroups_train = fetch_20newsgroups(subset='train', remove=('headers',␣
↪'footers', 'quotes'), categories=categories, random_state=42)
newsgroups_test = fetch_20newsgroups(subset='test', remove=('headers',␣
↪'footers', 'quotes'), categories=categories, random_state=42)
```

1.0.2 Data exploration

The target attribute is the integer index of the category:

[4]:
```
from collections import Counter
Counter(newsgroups_train.target)
```

[4]:
```
Counter({2: 594, 3: 593, 1: 584, 0: 480})
```

[5]:
```
Counter(newsgroups_test.target)
```

[5]:
```
Counter({2: 396, 3: 394, 1: 389, 0: 319})
```

Convert data to pandas dataframe

[6]:
```
train = pd.DataFrame({'text': newsgroups_train.data, 'labels': newsgroups_train.
↪target})
```

[7]:
```
print(len(train))
train.head(5)
```

```
2251
```

[7]:
```
                                                text  labels
0  WHile we are on the subject of the shuttle sof…       3
1  There is a program called Graphic Workshop you…       1
2                                                          2
3  My girlfriend is in pain from kidney stones. S…       2
4  I think that's the correct spelling..\n\tI am …       2
```

[8]:
```
test = pd.DataFrame({'text': newsgroups_test.data, 'labels': newsgroups_test.
↪target})
```

[9]:
```
print(len(test))
test.head(5)
```

```
1498
```

[9]:
```
                                                text  labels
0  \nAnd guess who's here in your place.\n\nPleas…       1
1  Does anyone know if any of Currier and Ives et…       1
2  =FLAME ON\n=\n=Reading through the posts about…       2
3  \nBut in this case I said I hoped that BCCI wa…       0
```

```
4  \nIn the kind I have made I used a Lite sour c…       2
```

Use a subset (10%) of the training set as a validation set

[10]:
```python
from sklearn.model_selection import train_test_split

train, dev = train_test_split(train, test_size=0.1, random_state=0,
                              stratify=train[['labels']])
```

[11]:
```python
print(len(train))
print("train:", train[['labels']].value_counts(sort=False))
train.head(3)
```

```
2025
train: labels
0         432
1         525
2         534
3         534
Name: count, dtype: int64
```

[11]:
```
                                                   text  labels
559   I wonder how many atheists out there care to s…       0
2060  We are interested in purchasing a grayscale pr…       1
1206  Dear Binary Newsers,\n\nI am looking for Quick…       1
```

[12]:
```python
print(len(dev))
print("dev:", dev[['labels']].value_counts(sort=False))
dev.head(3)
```

```
226
dev: labels
0         48
1         59
2         60
3         59
Name: count, dtype: int64
```

[12]:
```
                                                   text  labels
1570  I'd dump him.  Rude is rude and it seems he en…       2
1761  Hi Everyone ::\n\nI am  looking for  some soft…       1
455   A friend of mine has been diagnosed with Psori…       2
```

1.1 BERT

Define model's configuration

[13]:

```
# Model configuration # https://simpletransformers.ai/docs/usage/
↪#configuring-a-simple-transformers-model
model_args = ClassificationArgs()

model_args.overwrite_output_dir=True # overwrite existing saved models in the␣
↪same directory
model_args.evaluate_during_training=True # to perform evaluation while training␣
↪the model
# (eval data should be passed to the training method)

model_args.num_train_epochs=10 # number of epochs
model_args.train_batch_size=32 # batch size
model_args.learning_rate=4e-6 # learning rate
model_args.max_seq_length=256 # maximum sequence length
# Note! Increasing max_seq_len may provide better performance, but training␣
↪time will increase.
# For educational purposes, we set max_seq_len to 256.

# Early stopping to combat overfitting: https://simpletransformers.ai/docs/
↪tips-and-tricks/#using-early-stopping
model_args.use_early_stopping=True
model_args.early_stopping_delta=0.01 # "The improvement over best_eval_loss␣
↪necessary to count as a better checkpoint"
model_args.early_stopping_metric='eval_loss'
model_args.early_stopping_metric_minimize=True
model_args.early_stopping_patience=2
model_args.evaluate_during_training_steps=32 # how often you want to run␣
↪validation in terms of training steps (or batches)
```

With this configuration, the training will terminate if the eval_loss on the evaluation data does not improve upon the best eval_loss by at least 0.01 for 2 consecutive evaluations.

An evaluation will occur once for every 32 training steps.

[14]:
```
# Checking steps per epoch
steps_per_epoch = int(np.ceil(len(train) / float(model_args.train_batch_size)))
print('Each epoch will have {:,} steps.'.format(steps_per_epoch)) # 64 steps =␣
↪validating 2 times per epoch
```

```
Each epoch will have 64 steps.
```

Load the pre-trained model: model_type = bert; model_name = bert-base-cased (specifies the exact architecture and trained weights to use)

[16]:
```
model = ClassificationModel('bert', 'bert-base-cased', num_labels=4,␣
↪args=model_args, use_cuda=False) # CUDA is enabled
```

```
model.safetensors:   0%|          | 0.00/436M [00:00<?, ?B/s]
```

```
Some weights of BertForSequenceClassification were not initialized from the
```

```
model checkpoint at bert-base-cased and are newly initialized:
['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.
```

```
tokenizer_config.json:   0%|          | 0.00/49.0 [00:00<?, ?B/s]
```

```
vocab.txt:   0%|          | 0.00/213k [00:00<?, ?B/s]
```

```
tokenizer.json:   0%|          | 0.00/436k [00:00<?, ?B/s]
```

[17]:
```
print(str(model.args).replace(',', '\n')) # model args
```

```
ClassificationArgs(adafactor_beta1=None
 adafactor_clip_threshold=1.0
 adafactor_decay_rate=-0.8
 adafactor_eps=(1e-30
 0.001)
 adafactor_relative_step=True
 adafactor_scale_parameter=True
 adafactor_warmup_init=True
 adam_betas=(0.9
 0.999)
 adam_epsilon=1e-08
 best_model_dir='outputs/best_model'
 cache_dir='cache_dir/'
 config={}
 cosine_schedule_num_cycles=0.5
 custom_layer_parameters=[]
 custom_parameter_groups=[]
 dataloader_num_workers=0
 do_lower_case=False
 dynamic_quantize=False
 early_stopping_consider_epochs=False
 early_stopping_delta=0.01
 early_stopping_metric='eval_loss'
 early_stopping_metric_minimize=True
 early_stopping_patience=2
 encoding=None
 eval_batch_size=100
 evaluate_during_training=True
 evaluate_during_training_silent=True
 evaluate_during_training_steps=32
 evaluate_during_training_verbose=False
 evaluate_each_epoch=True
 fp16=False
 gradient_accumulation_steps=1
 learning_rate=4e-06
 local_rank=-1
 logging_steps=50
```

```
loss_type=None
loss_args={}
manual_seed=None
max_grad_norm=1.0
max_seq_length=256
model_name='bert-base-cased'
model_type='bert'
multiprocessing_chunksize=-1
n_gpu=1
no_cache=False
no_save=False
not_saved_args=[]
num_train_epochs=10
optimizer='AdamW'
output_dir='outputs/'
overwrite_output_dir=True
polynomial_decay_schedule_lr_end=1e-07
polynomial_decay_schedule_power=1.0
process_count=10
quantized_model=False
reprocess_input_data=True
save_best_model=True
save_eval_checkpoints=True
save_model_every_epoch=True
save_optimizer_and_scheduler=True
save_steps=2000
scheduler='linear_schedule_with_warmup'
silent=False
skip_special_tokens=True
tensorboard_dir=None
thread_count=None
tokenizer_name='bert-base-cased'
tokenizer_type=None
train_batch_size=32
train_custom_parameters_only=False
trust_remote_code=False
use_cached_eval_features=False
use_early_stopping=True
use_hf_datasets=False
use_multiprocessing=True
use_multiprocessing_for_evaluation=True
wandb_kwargs={}
wandb_project=None
warmup_ratio=0.06
warmup_steps=0
weight_decay=0.0
model_class='ClassificationModel'
labels_list=[0
```

```
1
2
3]
labels_map={0: 0
1: 1
2: 2
3: 3}
lazy_delimiter='\t'
lazy_labels_column=1
lazy_loading=False
lazy_loading_start_line=1
lazy_text_a_column=None
lazy_text_b_column=None
lazy_text_column=0
onnx=False
regression=False
sliding_window=False
special_tokens_list=[]
stride=0.8
tie_value=1)
```

Fine-tuning the model (takes a while)

[18]:
```
_, history = model.train_model(train, eval_df=dev)
```

```
  0%|          | 0/4 [00:00<?, ?it/s]
```

```
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
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false)
huggingface/tokenizers: The current process just got forked, after parallelism
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To disable this warning, you can either:
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false)
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
```

```
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
huggingface/tokenizers: The current process just got forked, after parallelism
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To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Epoch:   0%|          | 0/10 [00:00<?, ?it/s]

Running Epoch 1 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 2 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]
```

```
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 3 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 4 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
```

```
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 5 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 6 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]
```

```
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 7 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
```

```
Running Epoch 8 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 9 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
```

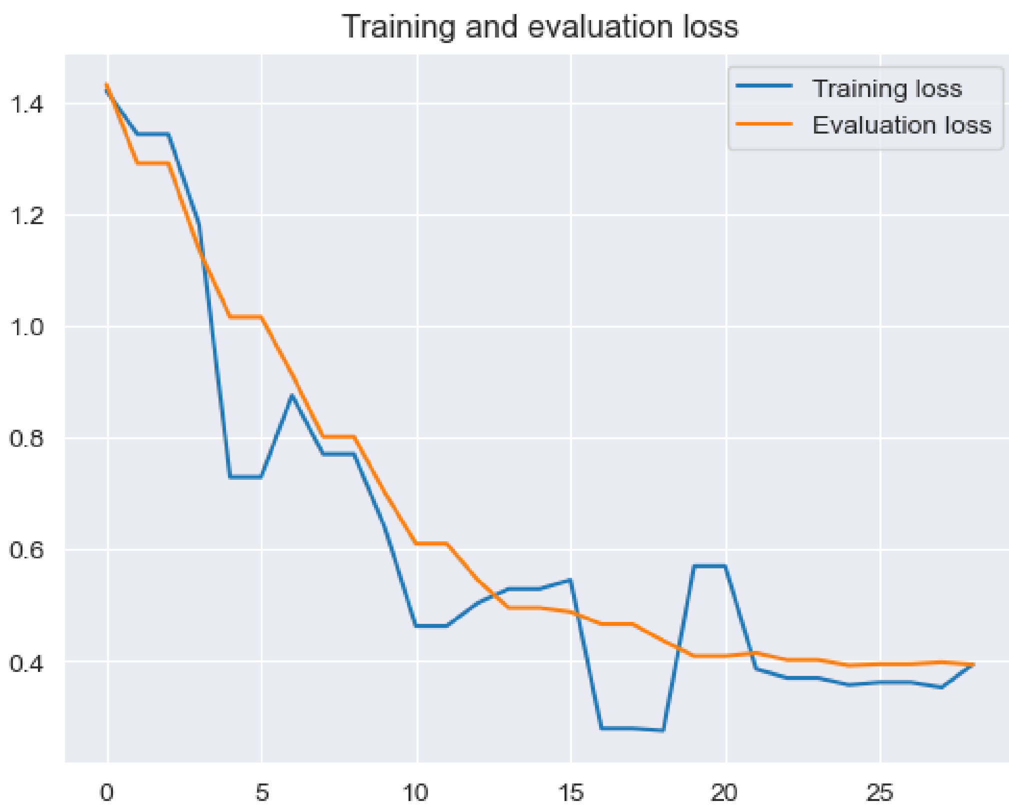
```
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

Running Epoch 10 of 10:   0%|          | 0/64 [00:00<?, ?it/s]

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)

0it [00:00, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
```

[19]:

```python
# Training and evaluation loss
train_loss = history['train_loss']
eval_loss = history['eval_loss']
plt.plot(train_loss, label='Training loss')
plt.plot(eval_loss, label='Evaluation loss')
plt.title('Training and evaluation loss')
plt.legend()
```

[19]:

```
<matplotlib.legend.Legend at 0x318109b20>
```



- Loss measures the "goodness" of your model
- The smaller the loss, the better the classifier is at modeling the relationship between the input data and the output targets
- But you need to be careful not to overfit

In our case, we stopped training because eval_loss loss did not improve upon the best eval_loss by at least 0.01 for 2 consecutive evaluations.

We can observe fluctuations in the training loss, but overall it is decreasing. We can have a smoother learning curve by varying hyperparameters, e.g., learning rate, batch size.

[20]:
```
# Evaluate the model
result, model_outputs, wrong_predictions = model.eval_model(dev)
result
```

```
0it [00:00, ?it/s]
```

```
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
```

```
false)

Running Evaluation:   0%|          | 0/3 [00:00<?, ?it/s]
```

[20]:
```
{'mcc': 0.8409281359739736, 'eval_loss': 0.39391950766245526}
```

- mcc: Matthews correlation coefficient
- eval_loss: Cross Entropy Loss for dev

Make predictions with the model (predict the labels of the documents in the test set)

[21]:
```
predicted, probabilities = model.predict(test.text.to_list())
test['predicted'] = predicted
```

```
  0%|          | 0/2 [00:00<?, ?it/s]

huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks…
To disable this warning, you can either:
        - Avoid using `tokenizers` before the fork if possible
        - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true |
false)
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  0%|          | 0/15 [00:00<?, ?it/s]
```

Test set predictions

[22]:
```
test.head(5)
```

[22]:

	text	labels	predicted
0	\nAnd guess who's here in your place.\n\nPleas…	1	1
1	Does anyone know if any of Currier and Ives et…	1	1
2	=FLAME ON\n=\n=Reading through the posts about…	2	2
3	\nBut in this case I said I hoped that BCCI wa…	0	0
4	\nIn the kind I have made I used a Lite sour c…	2	2

Evaluate the model's performance on the test set

[23]:
```python
# Result (note: your result can be different due to randomness in operations)
print(classification_report(test['labels'], test['predicted']))
```

```
              precision    recall  f1-score   support

           0       0.74      0.86      0.79       319
           1       0.88      0.91      0.89       389
           2       0.94      0.87      0.90       396
           3       0.88      0.79      0.84       394

    accuracy                           0.86      1498
   macro avg       0.86      0.86      0.86      1498
weighted avg       0.86      0.86      0.86      1498
```

1.1.1 End of this notebook.